

BUREAU OF INDIAN STANDARDS

NITS

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28.01.26

Sub: Guidelines for Allocation of Work to Interns Posted at BOs – 2026-27

This document is intended to serve as a guidance document for Branch Offices for allocation of work to interns being on-boarded for 8 weeks. The intent is to create awareness amongst the interns, being young students from different educational institutions, regarding the activities of BIS as well as to benefit from their energy and skill set to identify the gap areas, gain insights that will help in implementation and adoption of Indian Standards by the stakeholders.

Two interns of same discipline may be engaged in each BO. BOs may decide the disciplines considering the concentration of industries, products where frequency and number of non-compliances are high etc. Details of disciplines of interns may be obtained from the BOs every year. The plan for utilization of Interns in BOs has been categorized week wise as detailed below:

Introduction and Preparatory Work at BOs

1. Branch Offices should keep an updated industrial landscape of their jurisdiction to facilitate the visit of the interns to various units (manufacturing units, labs, AHCs). This would ensure efficient allocation based on their discipline upon posting. Utilizing the industrial landscape, a schedule of the units to be visited shall be prepared and the same shall be approved by Head BO. It is advisable that Head BO may have a prior interaction with the units to be visited so as to avoid infructuous visits. The BOs should identify, coordinate with, and brief the units on the intern's tasks prior to the visit.
2. BOs should also gather feedback from both the industry, the lab and AHCs regarding the intern's performance during the visit.
3. The 8 weeks internship at BOs should be broadly divided as follows:
 - Week 1-2: Orientation
 - Week 3-4: Visits to Industry
 - Week 5: Visit to Regional Lab & OSL
 - Week 6: Visit to AHC & virtual participation in BIS TC meeting
 - Week 7-8 : Summing up and Reporting
4. The final report of the intern shall be submitted by BO head to NITS along with the copy of the certificate (Annex 1), and a write-up covering satisfactory completion of the internship and details regarding benefits accrued to BIS as well as to the intern through this internship. Editable files of the Report and the findings shall be also submitted by BO Head to NITS with a copy to CMDs / concerned technical departments/ HMD for compiling the suggestions for improvements in standards, certification processes, laboratory and hallmarking activities.

Details of the activities to be carried out is as attached herewith:

Week 1-2: Orientation & Familiarization with BIS Activities

1.0 Interns will participate in a video conference (VC) hosted by NITS for an initial briefing on the 2nd half of the 1st day. In the first two weeks, the interns shall be oriented and educated regarding various BIS activities. The suggested module for Week1 and 2 is as given below:

	Week 1 – Orientation cont.
i)	Introduction to Indian Standards: Brief introduction to Standardization activity and Standard Formulation Process in BIS, Structure of Indian Standards, Scope, Raw material requirements, in-process controls in manufacturing, Performance and/or Safety requirements, Methods of Test, Marking, Packaging and labelling requirements etc.
ii)	Overview of BIS product certification: - Provisions of the BIS Act, 2016, BIS Rules, and the BIS (Conformity Assessment) Regulations, 2018, Voluntary and Compulsory certification with a focus on Scheme-I, QCOs, etc.
iii)	Overview of general and product specific guidelines: The guidelines of product certification activity and product manuals with a focus on market and factory surveillance, dealing with non-conformity, verification of corrective actions etc.
iv)	Introduction to IT tools deployed in certification: Introduction to Manakonline and use of apps for factory and market surveillance, Know your Standard module of BIS website, List of products under BIS compulsory certification, etc.
	Week 2 – Orientation cont.
i)	Operation of BIS License: Detailed discussions on post grant of license activities, review of performance of a license, sample non-conformity, root cause analysis, corrective actions, verification of corrective actions and post verification corrective actions by BIS.
ii)	Overview of Laboratory activities of BIS- Different BIS Labs, BIS recognized labs and Govt empanelled labs available, Basics of Lab recognition scheme, LIMS, test facilities available in our BIS Labs, etc.
iii)	Familiarization with the allocated product(s) through workshops/Q&A sessions: The BOs may organize Workshops/Q&A and technical sessions, to make the interns

	understand the product certification activity and product allocated to them for the subsequent 2 weeks, i.e. Week 3 and Week 4.
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Week 3-4: Visits to Industry

2.1	The interns shall visit Four manufacturing units for the products identified to make them understand the manufacturing processes and performance requirements and controls exercised to manufacture products meeting the quality requirements prescribed by Indian Standards and to avoid product non-compliance. While finalizing the units to be visited BOs are requested to select Two units from Large/Medium sector and Two units from Small/Micro sector so as to inform the interns about the sector wise differences. In case sufficient Large & medium units are not available then selection should include at least Two units demonstrating consistent compliance to requirements and Two which are lacking in the same. The visits to the industries may last more than one day, depending on the scale of the industry, as per the discretion of the Head of BO.
2.2	Industry visits should also incorporate demonstration on methods of test, understanding the relevance of tests, BIS factory surveillance mechanisms and verification of corrective actions. Interns shall understand the industry practice of root cause analysis and process of deciding and implementing corrective actions after non-conformity. The steps taken by manufacturers to deal with non-conformity and their effect on product quality are to discussed in detail during these visits.
2.3	During the interactions with manufacturers of different scale, the interns are expected to understand and record the difference in approach to manufacturing of product based on scale of factory. Key points for observations for the same are, composition of raw material, maintenance of incoming material quality, manufacturing processes deployed, technological difference, levels of automation, skill set of personnel, in-process controls, quality control, final product testing etc.. Interns should also interact with the industry with respect to the latest technological advancements in the sector and changes that industry envisage to keep abreast with the latest technological trends. Interns should obtain details about any additional variety/product/test methods that are currently missing in the Indian Standards.

	Interns should obtain details about the international standards being followed by the industry due to absence of Indian Standards or due to reasons/market compulsions, etc.
2.4	On visits to Industry the interns should explore opportunities for incorporating sustainability and resource efficiency in existing standards. They should study the incorporation of sustainable practices in manufacturing on the lines of but not limited to: ☐ Use of recyclable or eco-friendly materials. ☐ Energy-efficient processes. ☐ Circular economy initiatives etc..
2.5	The interns should utilize the factory visit opportunity for collecting information with respect to preparation of an analytical Report. The report should consolidate findings from visits and data analysis, compare practices of large, medium, small, and micro industries, highlight frequently observed non-conformities and corrective actions, document good practices for maintaining continuous compliance to standards and recommend their adoption, identify key technological and resource gaps, with solutions to bridge them.
2.6	The overall focus of interns during factory visits and subsequent reporting is to bring out any difficulties in implementation of Indian Standards by industry. Based on their field observation they are also expected to suggest improvements in existing BIS certification practices.

Note: Please see Annex 2 for suggestive industry visit Questionnaire

Week 5: Visit to Regional Laboratory and OSL

3.1	A two-day common visit to the concerned BIS regional laboratory is to be organized for all the interns posted in a Region. Heads of BOs in a region should coordinate with each other and the regional lab to schedule a common visit on a mutually agreed date, for all the interns posted in that region.
3.2	Another common visit of interns posted in a BO to an OSL(BIS Recognized lab) located within the jurisdiction of a BO is to be organized by the concerned Branch Office.

3.3	Visits to laboratories should focus on how laboratories work starting from receipt of samples at sample cell, basic concepts of testing including sample traceability, and calibration up to test report generation and uploading on LIMS.
3.4	The interns now familiar with product testing being undertaken by manufacturers, may contrast and compare the test methods, instruments and practices being used in factory and during independent testing of that product in laboratory.

Note: Please see Annex 3 for suggestive Questionnaire for laboratory visits

Week 6: Visit to Assaying and Hallmarking Centre & Virtual Participation in BIS Technical Committee meeting

4.0	A common visit of interns posted in a Branch Office to an AHC located within the jurisdiction of a BO is to be organized by the Branch Office.
4.1	During this visit interns will be required to understand the basics of Hallmarking Scheme for Gold and Silver jewellery, activities of a BIS recognized AHC, step-by step process of purity testing of gold through XRF and fire assay methods and of silver through gravimetric and/or potentiometric methods, laser marking of conforming articles, HUID.
4.2	Overview of Standard Formulation activity of BIS- List of Technical departments available and programme of work of those departments, Structure and Composition of a Technical department, Consensus based standard formulation process, Process of providing new work item proposals and comments on existing Indian Standards through BIS portal. The interns will also be required to virtually attend a meeting of BIS Technical Committee relevant to their academic discipline during this week. Interns and BO officers are requested to make advance preparation in this regard with concerned Member Secretary.

Note: Please see Annex 4 for suggestive Questionnaire for AHC visit

Weeks 7 and 8: Summing up and Reporting

5.0	In the 7 th week, a group discussion session shall be organized by the BO for sharing of views among the interns which shall be moderated by BO head or a senior officer.
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	During this discussion, based on their notes/records of their visits, each intern will be required to elaborate his or her observations vis-à-vis analysis of failures, corrective actions and manufacturing practices product-wise and industry wise. The discussions will also cover Good practices and Controls/checks deployed to ensure continuous compliance to Standards for the assigned product and observations noted during the Laboratory and AHC visit.
5.1	Based on this discussion, the interns will be required to prepare an analytical report recording aspects including but not limited to: a) The parameters in which various failures observed in allocated products, b) The manner and finding of root cause analysis and corrective action undertaken by different manufacturers for each failure, c) Effectiveness of implementation of corrective actions by the manufacturer, d) The steps taken to avoid recurrence of failures. e) The interns own technical assessment regarding the cause of failure and corrective actions being taken. f) Document good practices for maintaining continuous compliance to standards and recommend their adoption. g) Identify key technological and resource gaps, with solutions to bridge them. h) Suggest improvements in standards, emphasizing inclusivity, sustainability, and resource efficiency i) Any comments on the need for any parameters of the standards to be reviewed j) Any other aspect worth highlighting
5.2	The overall focus of reporting by interns to bring out any difficulties in implementation of Indian Standards by industry. Based on their field observation they are also expected to suggest improvements in existing BIS certification practices and/or Indian Standards.
5.3	This draft report covering the complete duration of internship shall be presented to the Head BO by the interns. The interns will prepare and make presentation individually for reporting their observations and findings. The Head BO's feedback on the report shall be incorporated and the finalized report shall be submitted by the interns at the end of week 8.
5.4	Along with the reports the interns will be required to prepare individual Presentations. The elements of the presentation to include: o key findings, insights, and recommendations.

	- o identification of good practices followed to ensure compliance to standards on allocated product and how other manufacturers can learn from compliant manufacturers' practices. - o Highlight actionable solutions for improving implementation of Indian Standards and - o improvements in existing BIS certification practices and/or Indian Standards.
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Other General Guidelines

- Total expenditure incurred on the intern to be also submitted by the BO Head to NITS in due course. Expenses would be met from the fund allocated under Budget Head 'Other Expenses - R&D project (Code 3104).
- The matters related to reimbursement of visit expenses, logistics facilities, payment of stipend and issuance of internship completion certificates, etc are to be dealt by BOs in accordance with Internship Scheme Guidelines issued by NITS.
- Web links for accessing various standards should be informed to the interns. Any Standard required by the interns for the study would be made available by the BO through BIS' Library/International Relations Department or BIS Portal (Know Your Standard)
- Period of internship can be extended by a week or so in case of genuine exigency subject to the approval by Head (BO).

ANNEX 1 FORMAT OF CERTIFICATE TO BE ISSUED TO INTERNS

BUREAU OF INDIAN STANDARDS XX BRANCH OFFICE

Ref: NITS/XXBO/INTERN/2026-27

DD.MM.YYYY

TO WHOMSOEVER IT MAY CONCERN

It is certified that Ms./Shri _____, Student of _____ has successfully completed the internship, during the period _to _, focusing on implementation of National Standards by the industry and analysis of user feedback, thereby contributing to the organization's understanding about the latest industry practices.

During the course of Internship his/her conduct was found to be satisfactory.

Head (BO)

ANNEX 2

Suggestive Internship Industry Visit Questionnaire

General Information:

1. Name and address of the Industry :
2. Date(s) of visit:
3. Type of Industry (e.g., manufacturing, service):
4. Size of the Industry (Large-scale unit / MSME):
5. Products manufactured:
6. Is the product intended for domestic market or export or both?

A. Raw Materials Used

1. What are the raw materials used?
2. Are Indian standards available for the raw materials. If no, what are the standards applicable for the raw materials (International standards/ company standards or any other)
3. How is the quality and consistency of raw materials ensured? (ISI marked / any other certification / supplier certificate / inhouse tests/ test at outside laboratories)
4. Any vendor approval has been carried out? Procedure for vendor selection.
5. Actions taken on non-complying raw material.
6. Any precautions taken for storage of raw material.

B. Manufacturing Processes

1. Describe the overall production process for the primary products/services.
2. Whether production process is batch / continuous? How is traceability of the product ensured
3. Are there standard operating procedures (SOPs) for all key processes in your company? Can you describe one or two important SOPs?
4. What are the Critical Control points and the control parameters , method of in process controls and frequency of checks?
5. Automation if any. If Yes How is automation integrated into the manufacturing process?
6. Are there any customizations and/or specific challenges in the production process?
7. What are the Good Manufacturing Practises followed?

C. Testing

1. What are the tests carried out?
2. Does Indian Standard exist for test methods for each parameter?
3. Are the tests carried out as per Indian Standards? If no, which standard is followed in testing. (mention separately for each parameter)
4. If International Standards are being followed, then reasons for the same. Also difference in Indian and International Standards to be highlighted.
5. Any additional tests carried out other than those specified in the Indian standard, If Yes, then state the purpose of such tests.
6. Frequency of tests carried out.
7. Any Automated test equipment used, if yes the details of the equipment including name of manufacturer, brochure and tests for which it is used.

8. Are test facilities for all parameters available in-house? Parameters for which in-house test facility not available.
9. Details of outside labs where test facilities are available

D. Non-Conformities and Process Improvements

1. What are the most common non-conformities observed in the product.
2. What are the root causes of these non-conformities?
3. Details of preventive actions to ensure that such non conformities are not repeated.
4. Actions taken on non-conforming products.
5. What tools or methods (e.g., Six Sigma, Lean) are used for continuous improvement?
6. Any recent case study where non compliances have been reduced and improvements have been made in the product quality.

E. Sustainability Practices

1. How does the company integrate sustainability in their production process (e.g., resource optimization, waste reduction, any other)
2. What are the main sustainable practices followed in sourcing raw materials and managing resources?
3. How is the waste generated during production managed.
4. Are there any practices in place to reuse or recycle materials or products within production cycle. If so, what is the percentage.
5. What are the impacts of the production process on environment and measures taken by the firm in environmental protection.

F. Energy Efficiency

1. How is energy consumption managed and monitored during the manufacturing process
2. Does the firm have any initiative in place to reduce energy consumption, including use of renewable energy. If so details of the same
3. Any energy audits or monitoring systems to assess energy usage

G. Corporate Social Responsibility (CSR)

1. Details of CSR initiatives undertaken, if any (e.g., community development, environmental sustainability)
2. Details of engagement with local communities and stakeholders to create a positive social impact
3. How does the firms CSR activities align with their core values and business goals.

Note: BOs may customize the questionnaire based on their specific requirements.

ANNEX 3

Suggestive Questionnaire for Laboratory visits (Both for Regional lab and OSL)

General Information :

1. Name and address of the Laboratory visited.
2. Date(s) of visit
3. Type of lab (e.g., BIS Lab/BIS recognized lab/Govt empanelled lab):
4. Sections available in the lab based on the product category tested:
5. Testing of products witnessed and IS number and parameters

A. Testing

1. Explain the role of this laboratory in verifying manufacturers' compliance with Indian Standards and supporting BIS certification and surveillance activities.
2. Tell us in details regarding the process of receipt of samples at sample cell to test report generation and uploading on LIMS.
3. What were the critical control points ensuring integrity and traceability of the sample?
4. For the products you have seen in factory visits, how do the test methods followed in the laboratory align with the relevant Indian Standards? Did you observe any challenges in interpretation or execution of these methods?
5. Are the test methods in Indian Standards aligned with international standards? If not, do you see any scope for harmonization or upgradation of Indian test methods based on international practices observed? Provide reasons.
6. What are the latest testing technologies or instruments available in the market (even if not presently installed) that can significantly improve efficiency, turnaround time, or testing coverage?
7. Why is calibration critical for laboratory credibility? What risks may arise from improper calibration? How is calibration frequency decided?
8. What aspects of laboratory practices inspired confidence in test results, and what improvements (if any) would you suggest based on your academic understanding?
9. How did this laboratory visit enhance your understanding of the role of independent testing in enforcement of Indian Standards?
10. Summarize the three most important technical or procedural learnings from the laboratory visit that will help you better understand BIS certification, surveillance, or standard implementation.

Note: BOs may customize the questionnaire based on their specific requirements.

ANNEX 4

Suggestive Questionnaire for AHC visit

General Information :

- a. Name and address of the AHC visited.
- b. Date(s) of visit
- c. Scope of recognition of the AHC (only yellow gold artefact or Both yellow gold and silver artefacts) and the grades.

A. Hallmarking activities

1. What are the different methods of sampling and the tools required for the same?
2. What is the step-by-step process of determining the amount of gold (in ppt) in any article?
3. How many XRF readings are to be taken for each sample?
4. Which points in a sample should be essentially considered for obtaining the XRF readings?
5. What is the step-by-step process of determining the amount of silver (in ppt) in any article? (if it is recognized for silver artefacts also)
6. What is HUID? What is the process of laser marking on conforming articles?
7. List of equipment and reagents/chemicals required in fire assay process.
8. What are the prohibited metals in a gold article as per IS 1417?
9. What safety precautions are followed while handling acid, lead?

Note: BOs may customize the questionnaire based on their specific requirements.